

Lemma 3.2: The map described above is surjective.

Proof:

Let $w \in T_a \mathbb{R}^n$ be any derivation. Then having let $e_1, \dots, e_n$ be the standard basis, for all $1 \leq i \leq n$ let $c_i := w(x \mapsto x \cdot e_i)$. I claim $w = D_v|_a$ where $v = c_1 e_1 + \dots + c_n e_n$.

To see why, let f be any C^∞ function on $\mathbb{R}^n$. Next for any $x \in \mathbb{R}^n$ let $h(t) := f(a + t(x - a))$. If we say $a = (a_1, \dots, a_n)$ and $x = (x_1, \dots, x_n)$, then note that:

$$\begin{aligned} f(x) - f(a) &= h(1) - h(0) = \int_0^1 \frac{dh}{dt}(s) ds \\ &= \int_0^1 (x - a) \cdot \nabla f(a + s(x - a)) ds \\ &= \sum_{i=1}^n (x_i - a_i) \int_0^1 \frac{\partial}{\partial x_i} f(a + s(x - a)) ds. \end{aligned}$$

In particular, we set $h_i(x) := \int_0^1 \frac{\partial}{\partial x_i} f(a + s(x - a)) ds$. By Leibniz's rule applied h_i (which we can apply to x in bounded subsets of $\mathbb{R}^n$ since f is a C^∞ function), we know h_i is also a C^∞ function on $\mathbb{R}^n$.

As a side note, if f were only finitely many times differentiable then this paragraph would not work. This hints at how we need an alternate definition of tangent vectors for when we are working on C^r manifolds where $r < \infty$.

And now we have that $w(f) = w(f(a) + \sum_{i=1}^n (x_i - a_i)g_i(x))$. Yet $w(f(a)) = 0$ since $f(a)$ is a constant. Meanwhile:

$$\begin{aligned} w((x_i - a_i)g_i(x)) &= (a_i - a_i)w(g_i) + g_i(a)w(x_i - a_i) \\ &= 0 + (w(x \mapsto x \cdot e_i) - w(a_i)) \int_0^1 \frac{\partial}{\partial x_i} f(a) ds = (c_i - 0) \frac{\partial}{\partial x_i} f(a). \end{aligned}$$

So, $w(f) = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i} f(a) = D_v|_a f$. ■

As a side note, if we consider "derivations" whose domain is the vector space $C^k(M, \mathbb{R})$ where M is a C^r manifold and $k < \min(r + 1, \infty)$, then we can ask about "exotic derivations" which aren't described by directional derivatives.

Corollary: A vector space basis for $T_a \mathbb{R}^n$ is given by the partial derivative operators $\frac{\partial}{\partial x_i}$ at a .

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Lemma 3.11: Suppose $\iota : H^n \hookrightarrow \mathbb{R}^n$ is the inclusion map. (Hopefully it is obvious that this is a C^∞ -map.) For any $a \in \partial H^n$ the differential $d\iota_a : T_a H^n \rightarrow T_a \mathbb{R}^n$ is an isomorphism.

Proof:

First I shall show injectivity. Suppose $v \in T_a H^n$ and $d\iota_a(v) = 0 \in T_a \mathbb{R}^n$. Then consider any smooth function $f : H^n \rightarrow \mathbb{R}$. By definition we know that f extends to a smooth function $g : U \rightarrow \mathbb{R}$ where $U \subseteq \mathbb{R}^n$ is open and contains H^n . Furthermore, H^n is a closed subset of U . Thus by identical reasoning as in the proof of proposition 3.9, we can find a smooth function $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $\tilde{f}|_{H^n} = g|_{H^n} = f$.

But now $v(f) = v(\tilde{f}|_{H^n}) = v(\tilde{f} \circ \iota) = d\iota_a(v)\tilde{f} = 0$. And since f was arbitrary, we've proven that $v = 0$. So, $d\iota_a$ has a trivial kernel and is thus injective.

Next we show surjectivity. Suppose $w \in T_a \mathbb{R}^n$ and then define $v \in T_a H^n$ by $v(f) := w(\tilde{f})$ where $\tilde{f}$ is any smooth extension of f to all of $\mathbb{R}^n$.

To see that v is well-defined, note by the prior corollary that $w = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i} \big|_a$ for some $c_1, \dots, c_n \in \mathbb{R}$. Yet since $\tilde{f}$ has a continuous derivative, $\frac{\partial}{\partial x_i} \tilde{f}(a)$ is uniquely determined by $\tilde{f}$'s restriction to H^n .

Now its easy to check that v is a derivation and that $d\iota_a(v) = w$. ■

Proposition 3.12: Given any n -dimensional C^∞ manifold M with or without boundary, for all $p \in M$ we have that $T_p M$ is an n -dimensional vector space.

Proof:

We already know from [page 1034](#) that $T_p M$ is isomorphic to $T_a \hat{U}$ where $\hat{U}$ is some open subset of H^n or $\mathbb{R}^n$ and $a \in \mathbb{R}^n$. But now it's straightforward to prove that $T_a \hat{U}$ is isomorphic to $T_a \mathbb{R}^n$ by just chaining together the isomorphisms induced by varying inclusion functions. And finally, [lemma 3.2](#) tells us that $T_a \mathbb{R}^n$ is n -dimensional. Thus, $T_p M$ is also n -dimensional. ■

I want to switch to covering some category theory related notes about manifolds.

Firstly, we let **Man** be the category whose objects are topological manifolds without boundaries and whose morphisms are continuous maps. If we want to allow our manifolds to have a boundary, we instead consider the category **Man_b**.

Proposition: Suppose $M_1, \dots, M_{k-1}$ are topological manifolds without boundary and M_k is a topological manifold with boundary. Also suppose each M_i has dimension n_i . Then the topological product space $M := M_1 \times \dots \times M_k$ is an $n_1 + \dots + n_k$ dimensional topological manifold with boundary.

Proof:

Countable products of second countable topological spaces are still second countable. Hence M is second countable.

This is easy to verify.

Also arbitrary products of Hausdorff spaces are still Hausdorff. Hence, M is Hausdorff.

Finally, to show M is locally Euclidean, let $(p_1, \dots, p_k) \in M$. Then for each $1 \leq i \leq k$ we can pick a chart (U_i, φ_i) on M_i containing p_i . Now we identify $\mathbb{R}^{n_1} \times \dots \times \mathbb{R}^{n_k}$ with $\mathbb{R}^{n_1 + \dots + n_k}$ in the obvious way. Then we consider the chart $(U_1 \times \dots \times U_k, \varphi_1 \times \dots \times \varphi_k)$. This will be a homeomorphism to an open subset of $\mathbb{R}^{n_1 + \dots + n_k}$ or $H^{n_1 + \dots + n_k}$. ■

Remark: By the proposition on [pages 1010-1011](#), we know that $(p_1, \dots, p_k)$ is a boundary point of M if and only if p_k is a boundary point of M_k . Hence if M_k has an empty boundary then our constructed chart from before will always be an interior chart instead of a boundary chart.

Since $M = M_1 \times \dots \times M_k$ is also a product in the category of topologies, we know M interacts with morphisms (i.e. continuous maps) in the way that a category theoretic product should. And by the above remark, we've thus proven that **Man** admits finite products.

As for whether $\mathbf{Man}_b$ admits finite products, we can actually assume in the prior proposition that $M_1, \dots, M_{k-1}$ have boundaries. To see why, let:

$$\mathbb{R}_+^k := \{(x_1, \dots, x_k) : x_i \geq 0 \text{ for all } i\}.$$

Proposition: $\mathbb{R}_+^k$ is homeomorphic to H^k .

Proof:

Firsty, note that $\mathbb{R}_+^1 = H^1$. Also $\mathbb{R}_+^2$ is homeomorphic to H^2 via the map:
 $\varphi(x, y) := (x^2 - y^2, 2xy)$.

Why?

It's clear that φ is continuous and also that it maps $\mathbb{R}_+^2$ into H^2 . After all, $x \geq 0$ and $y \geq 0$ implies that $2xy \geq 0$. As for showing the other requirements for φ to be a homeomorphism, I want to explicitly solve for the inverse function φ^{-1} .

Suppose $\varphi(x, y) = (z, w)$.

Suppose $y \neq 0$. Then $w = 2xy \implies x^2 = \frac{w^2}{4y^2}$. In turn, we get that
 $z = x^2 - y^2 = \frac{w^2}{4y^2} - y^2$. Or in other words, $4y^4 + 4zy^2 - w^2 = 0$. By using the quadratic formula, we can then solve that:

$$y^2 = \frac{-4z \pm \sqrt{16z^2 + 16w^2}}{8} = \frac{-z \pm \sqrt{z^2 + w^2}}{2}$$

We know $y^2 \geq 0$ though. So, we can actually further solve that:

$$y^2 = \frac{-z + \sqrt{z^2 + w^2}}{2}$$

And then since y is nonnegative, we can finally say that $y = \sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}}$.

And this is a well-defined real number because $\sqrt{z^2 + w^2} \geq \max(|-z|, 0)$.

Also, since $y \neq 0$ by assumption we can say that:

$$x = \frac{w}{2y} = \frac{w}{2\sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}}}.$$

So, we define: $\psi : H^2 - \{(z, 0) : z \geq 0\} \rightarrow \mathbb{R}_+^2 - \{(x, 0) : x \geq 0\}$ by:

$$\psi(z, w) = \left(\frac{w}{2\sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}}}, \sqrt{\frac{-z + \sqrt{z^2 + w^2}}{2}} \right)$$

Clearly ψ is a continuous function. And by how we solved for ψ we know that $\psi(\varphi(x, y)) = (x, y)$. Similarly, we can solve that $\varphi(\psi(z, w)) = (z, w)$.

Finally, note that if $y = 0$ then $w = 0$ and $z = x^2$. Therefore, we want to define $\psi(z, 0) := (\sqrt{z}, 0)$ when $z \geq 0$. And once we define this, we know that $\psi = \varphi^{-1}$. That said, we need to show our extension of ψ to the ray $\{(z, 0) : z \geq 0\}$ is still continuous.

To do this, consider any $z \geq 0$ and pick $R > \sqrt{z}$. Then $N := \overline{B_R(0)} \cap \mathbb{R}_+^2$ is a compact neighborhood of $\varphi^{-1}(z, 0)$ relative to $\mathbb{R}_+^2$. Also, we can see that:

$$\varphi(N) = \varphi(\overline{B_R(0)} \cap \mathbb{R}_+^2) = \overline{B_{R^2}(0)} \cap H^2 =: N'.$$

This is because $|\varphi(x, y)| = |(x, y)|^2$ and we already know from earlier that φ surjectively maps $\mathbb{R}_+^2$ into H^2 . So, φ is a bijective continuous map from the compact neighborhood N of $\varphi^{-1}(z, 0)$ onto a compact neighborhood N' of $(z, 0)$. It follows from the compactness of N and the Hausdorffness of N' that φ^{-1} is a continuous map from N' to N . In other words, φ^{-1} is continuous at $(z, 0)$.

But now its easy to conclude by induction that $\mathbb{R}_+^k$ is homeomorphic to H^k . After all, $\mathbb{R}_+^k = \mathbb{R}_+^{k-2} \times \mathbb{R}_+^2$ is homeomorphic to $\mathbb{R}_+^{k-2} \times H^2$ by the prior reasoning, and then that is trivially homeomorphic to $\mathbb{R} \times \mathbb{R}_+^{k-1}$. And now we just conclude by induction that $\mathbb{R}_+^{k-1} \cong H^{k-1}$. So, $\mathbb{R} \times \mathbb{R}_+^{k-1} \cong \mathbb{R} \times H^{k-1} \cong H^k$. ■

Corollary: Suppose $M_1, \dots, M_k$ are topological manifolds with boundary. Also suppose each M_i has dimension n_i . Then the topological product space $M := M_1 \times \dots \times M_k$ is an $n_1 + \dots + n_k$ dimensional topological manifold with boundary.

Proof:

Second countability and Hausdorffness is obvious. Hence, we only need to worry about if M is locally Euclidean.

Let $(p_1, \dots, p_k)$ be any point in M and let (U_i, φ_i) be a chart in M_i containing p_i . By shrinking each U_i , we can assume without loss of generality that $\widehat{U}_i := \varphi(U_i)$ is a rectangle in H^{n_i} whose ends are all open intervals in $\mathbb{R}$ or $[0, \infty]$. Then, we consider the function $\varphi_1 \times \dots \times \varphi_n : U_1 \times \dots \times U_k \longrightarrow \widehat{U}_1 \times \dots \times \widehat{U}_k$. By composing this function with some permutation of our coordinates, we get that there are open sets $V \subseteq \mathbb{R}^{n_1 + \dots + n_k - k}$ and $W \subseteq \mathbb{R}_+^k$, as well a homeomorphism $\psi : U \rightarrow V \times W$. Finally, transform W into an open set of H^k via a homeomorphism we know exists from the prior proposition. ■

Next, given any $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$ we let **Diff**^{*r*} denote the category whose objects are manifolds without boundaries and that are equipped a C^r -structure. The morphisms of **Diff**^{*r*} are C^r maps between manifolds. Also if we want to allow our manifolds to have boundaries then we consider the category **Diff**_{*b*}^{*r*}.

Proposition: Suppose $M_1, \dots, M_{k-1}$ are C^r manifolds without boundary and M_k is a C^r manifold with boundary. Then there is a canonical product C^r -structure we can equip $M := M_1 \times \dots \times M_k$ with to turn M into a C^r manifold.

Proof:

Given any smooth charts (U_i, φ_i) on M_i for $1 \leq i \leq k$ let $(U_1 \times \dots \times U_k, \varphi_1 \times \dots \times \varphi_k)$ be a smooth chart on M . Hopefully it's clear that these charts cover all of M . Also, suppose $(V_1 \times \dots \times V_k, \psi_1 \times \dots \times \psi_k)$ is another smooth chart on M of this form. Then the transition function between our charts would be given by:

$$(\psi_1 \circ \varphi_1^{-1}) \times \dots \times (\psi_k \circ \varphi_k^{-1}).$$

And since ψ_i and φ_i are C^r -compatible for each i , we know our transition function is a C^r -map. So, $(U_1 \times \dots \times U_k, \varphi_1 \times \dots \times \varphi_k)$ and $(V_1 \times \dots \times V_k, \psi_1 \times \dots \times \psi_k)$ are C^r -compatible.

Now to finish, just take the maximal atlas containing all the charts we just named. ■

Proposition: Suppose $M_1, \dots, M_k$ are C^r manifolds and at most one of the M_i has a nonempty boundary. Then let $M := M_1 \times \dots \times M_k$ and equip M with the product structure.

- The projection map $\pi_i : M \rightarrow M_i$ is a C^r map.

Why?

Given any $(p_1, \dots, p_k) \in M$, we can pick smooth charts (U_j, φ_j) on M_i containing p_j for every j . And now, upon setting $U := U_1 \times \dots \times U_k$ and $\varphi := \varphi_1 \times \dots \times \varphi_k$, we have that (U, φ) is a smooth chart on M and (U_i, φ_i) is a smooth chart on M_i such that $(p_1, \dots, p_k) \in U$ and $\pi(U) \subseteq U_i$. Finally, let $V_j := \varphi_j(U_j)$ for all j . Then $\varphi_i \circ \pi_i \circ \varphi^{-1}$ is easily checked to just be the projection map from $V_1 \times \dots \times V_k$ onto V_i . Since that is a C^∞ -map in the real-analysis sense, we can conclude π_i is a C^r map.

- The inclusion map $\iota_i : M_i \rightarrow M$ is a C^r map.

I'm omitting the proof of this cause its similar to the proof of the prior bullet point.

- Suppose N is a C^r -manifold with or without boundary. Then given any C^k maps $F_i : N \rightarrow M_i$ (where $k \leq r$), we have that $F : N \rightarrow M$ defined by $F(p) = (F_1(p), \dots, F_k(p))$ is a C^k -map from $N \rightarrow M$.

Why?

Consider any $p \in N$. Since each F_i is a C^k map, we know for each i that there is a smooth chart $(U^{(i)}, \varphi^{(i)})$ on N containing p as well as a smooth chart (V_i, ψ_i) on M_i such that $F_i(U^{(i)}) \subseteq V_i$ and $\psi_i \circ F_i \circ (\varphi^{(i)})^{-1}$ is a C^k function. By letting $U := U^{(1)} \cap \dots \cap U^{(k)}$ and letting φ be any of the $\varphi^{(i)}$ priorly chosen, we thus get a single smooth chart (U, φ) on N such that $F_i(U) \subseteq V_i$ and $\psi_i \circ F_i \circ \varphi^{-1}$ is a C^k map for all i .

Now let $\psi := \psi_1 \times \dots \times \psi_k$ and $V := V_1 \times \dots \times V_k$. Then (V, ψ) is a smooth chart on M such that $F(U) \subseteq V$. Plus $\psi \circ F \circ \varphi^{-1}$ is precisely the map $x \mapsto (\psi_1 \circ F_1 \circ \varphi^{-1}(x), \dots, \psi_k \circ F_k \circ \varphi^{-1}(x))$. And since all the components of this function are C^k functions, we know the entire thing is a C^k function. So, F is a C^k map from N to M .

Remark: This also proves that M along with the π_i is a category theoretic product. After all, F is the unique morphism from M to N with $\pi_i \circ F = F_i$ for each i when we forget the manifold structures on everything and just look at M, N , and the M_i as sets.

- Conversely to the last bullet point, suppose $F : N \rightarrow M$ is a C^k map (where $k < r$). Then let $F_i := \pi_i \circ F$ for each i . Since each π_i is a C^r map, we know that each F_i is a C^k map. Also $F(p) = (F_1(p), \dots, F_k(p))$. ■

In particular, the prior two propositions show that $\mathbf{Diff}^r$ admits products over any finite collection of objects, and also that $\mathbf{Diff}_b^r$ admits products over finite collections of objects satisfying special conditions.

In general though, we'd have to consider C^r manifolds with corners in order to take products over arbitrary finite collections of objects in $\mathbf{Diff}_b^r$.

In an n -dimensional topological manifold with corners M is defined to be a second countable Hausdorff space such that every $p \in M$ has an open neighborhood U that is homeomorphic to an open subset V of $\mathbb{R}^n$, H^n , or $\overline{\mathbb{R}_+^n}$.

Since $\overline{\mathbb{R}_+^n}$ is homeomorphic to H^n , we can say that M is an n -dimensional topological manifold with corners if and only if M is an n -dimensional topological manifold with a boundary.

Now suppose $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, U is an open subset of $\overline{\mathbb{R}_+^n}$, and $f \in C^r(U)$. Then we know there is an open set $V \subseteq \mathbb{R}^n$ containing U and a C^r map $g \in C^r(V)$ such that $g|_U = f$. Yet by identical reasoning as on [pages 83-84](#), we can then say that $\partial^\alpha g(x)$ is uniquely determined by f for all $x \in \partial\overline{\mathbb{R}_+^n}$ and multi-indices α with $|\alpha| \leq r$. Hence, we can uniquely refer to f as having derivatives on $U \cap \partial\overline{\mathbb{R}_+^n}$.

Consequently from the prior paragraph, we can now talk about a corner chart on M (i.e. a chart (U, φ) with U open in $\overline{\mathbb{R}_+^n}$ but not H^n or $\mathbb{R}^n$) being C^r compatible with another chart on M . And then, we can consider a maximal atlas on M containing some atlas of C^r -compatible charts (some of which potentially may be corner charts). As a side note, the proposition on [pages 142-143](#) still holds by identical reasoning.

We say a topological manifold with corners M equipped with a maximal C^r atlas (a.k.a a C^r structure) is a C^r -manifold with corners. Also we shall still call the charts in that structure "smooth charts".

Supposing M is a C^r -manifold with corners, we call $p \in M$ a corner point if there exists a smooth corner chart (U, φ) on M containing $p \in U$ such that at least two coordinates of $\varphi(p)$ in U are zero.

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Before doing anything else today, I want to go further in depth into the claim that $\overline{\mathbb{R}_+^n}$ is homeomorphic to H^n .

Recall from [pages 1037-1038](#) that $\varphi(x, y) := (x^2 - y^2, 2xy)$ is a homeomorphism between $\overline{\mathbb{R}_+^2}$ and H^2 . But then importantly, we also have that $(x, y) \mapsto (x, \frac{1}{2}y)$ is a homeomorphism between H^2 and H^2 . Therefore, $\tilde{\varphi}(x, y) := (x^2 - y^2, xy)$ is also a homeomorphism from $\overline{\mathbb{R}_+^2}$ and H^2 . And now by induction we can explicitly conclude that the map ψ defined by:

$$(x_1, \dots, x_k) \mapsto (x_1^2 - x_2^2, x_1^2 x_2^2 - x_3^2, \dots, (\prod_{i=1}^{n-1} x_i^2) - x_n^2, \prod_{i=1}^n x_i)$$

is a homeomorphism from $\overline{\mathbb{R}_+^n}$ to H^n .

One particular thing I want to point out is that $\psi(\partial\overline{\mathbb{R}_+^n}) = \partial H^n$.

One may notice that the homeomorphism I specifically wrote before is also a smooth map from $\mathbb{R}_+^n$ into H^n . A simple corollary of the following theorem will let us conclude that the function's inverse won't be differentiable from H^n into $\mathbb{R}_+^n$ however.

Theorem: Suppose M is a n -dimensional C^r -manifold with corners and that $p \in M$ is a corner point due to the smooth chart (U, φ) . Then for any smooth chart (V, ψ) on M containing p , we have that (V, ψ) is also a corner chart and at least two coordinates of $\psi(p)$ in V are zero.

Proof:

To start off, we can't have that $\psi(V)$ is open in $\mathbb{R}^n$. After all, if not then the function $\varphi \circ \psi^{-1}$ would violate the invariance of domain theorem (see [page 1008](#)). In a similar vein, we can't have that $\psi(V)$ is open in H^n or $\overline{\mathbb{R}_+^n}$ and also have that $\psi(p)$ is not on the boundary of those sets. After all, we'd otherwise be able to shrink V to an open subset of $\mathbb{R}^n$. So there are exactly two possibilities. Either:

- $\psi(V)$ is open in H^n but not $\mathbb{R}^n$ and $\psi(p) \in \partial H^n$,
- $\psi(V)$ is open in $\overline{\mathbb{R}_+^n}$ but not H^n or $\mathbb{R}^n$ and $\psi(p) \in \partial \overline{\mathbb{R}_+^n}$.

We'll first focus on disproving the first possibility. So, suppose for the sake of contradiction that $\psi(V)$ is open in H^n but not $\mathbb{R}^n$ and that $\psi(p) \in \partial H^n$.

Let $\hat{U} = \varphi(U)$ and without loss of generality suppose that:

$$\varphi(p) = (0, 0, x_3, \dots, x_n).$$

Note that we can always permute our coordinates to move two of the zero coordinates of $\varphi(p)$ to the first indices. After all, a coordinate permutation function P is a smooth diffeomorphism of $\overline{\mathbb{R}_+^n}$ to itself. Hence, the chart $(U, P \circ \varphi)$ will be C^r compatible with any other chart (V, ψ) since $P \circ (\varphi \circ \psi^{-1})$ and $(\psi \circ \varphi^{-1}) \circ P^{-1}$ are both C^r maps.

Next let $e_1, \dots, e_n$ be the standard basis for $\mathbb{R}^n$ and then consider the hyperplane $S := \{y \in \mathbb{R}^n : y \cdot e_n = 0\}$. Importantly, we know that $S' := S \cap \psi(U \cap V)$ is an open subset of S containing $\psi(p)$. Also let $\alpha = \varphi \circ \psi^{-1}$. Then since α is a C^r function from $\psi(U \cap V)$ to $\varphi(U \cap V)$ with a C^r inverse function, we know that $D\alpha$ must be an invertible linear map at all $y \in \psi(U \cap V)$. In particular, this means that $T := (D\alpha(\psi(p))) (S)$ is an $(n - 1)$ -dimensional subspace of $\mathbb{R}^n$.

But now consider that there must exist $v = (v_1, \dots, v_n) \in T$ such that either $v_1 \neq 0$ or $v_2 \neq 0$. After all, if not then T would be contained in an $(n - 2)$ -dimensional subspace of $\mathbb{R}^n$. By replacing v with $-v$ if needed, we can assume that either $v_1 < 0$ or $v_2 < 0$.

Finally, let $\gamma : (-\varepsilon, \varepsilon) \rightarrow S'$ be any C^1 curve such that $\gamma(0) = \psi(p)$ and $(\alpha \circ \gamma)'(0) = (D\alpha(\psi(p)))\gamma'(0) = v$. But now for small enough $t \in (0, \varepsilon)$, we must have that $(\alpha \circ \gamma)(t) \cdot e_1 < 0$ or $(\alpha \circ \gamma)(t) \cdot e_2 < 0$. And this contradicts that α maps $S' \subseteq \psi(U \cap V)$ into $\varphi(U \cap V) \subseteq \overline{\mathbb{R}_+^n}$.

With that, we can now conclude that $\psi(V)$ is open in $\overline{\mathbb{R}_+^n}$ but not H^n or $\mathbb{R}^n$, and also we must have that $\psi(p) \in \partial \overline{\mathbb{R}_+^n}$. In particular, if $\psi(p) = (y_1, \dots, y_n)$ that at least one y_i

must equal 0. To finish off, we need to prove another coordinate y_j equals 0 (where $j \neq i$).

Suppose otherwise. Then after permuting the coordinate of ψ we can assume without loss of generality that $\psi(p) = (y_1, \dots, y_{n-1}, 0)$ where $y_j \neq 0$ for all $1 \leq j \leq n-1$. Yet now we can shrink V such that $p \in V$ still and also $\psi(V)$ is open in H^n .

For example, restrict V to the set $V \cap \psi^{-1}(((0, \infty)^{n-1} \times [0, \infty)) \cap \psi(V))$.

And now we've contradicted the earlier conclusion in our proof. ■

We can define what it means for a map F from a manifold with corners M to another manifold N to be a C^r -map identically to how we defined C^r maps from manifolds without corners. And everything on [pages 146-151](#) still works via identical reasoning. Furthermore, the proof of the lemma on [pages 1031-1032](#) still works for manifolds with corners when we just replace every instance of H^n with $\overline{\mathbb{R}_+}^n$. And once we have that lemma, we can construct partitions of unity and prove Urysohn's lemma without even acknowledging that M has corners.

One more related note is that if M is a C^∞ manifold with corners, then for any $p \in M$ we can define derivations at p , the vector space $T_p M$, and the differential dF_p of a C^∞ function $F : M \rightarrow N$ at p identically to how we priorly did so when M was a manifold without corners.

By analogous reasoning as in lemma 3.11 on [pages 1035-1036](#), we can show that if $\iota : \overline{\mathbb{R}_+}^n \hookrightarrow \mathbb{R}^n$ is the inclusion map then $d\iota_a : T_a \overline{\mathbb{R}_+}^n \rightarrow T_a \mathbb{R}^n$ is an isomorphism for all $a \in \partial \overline{\mathbb{R}_+}^n$. And it thus follows that every fact about C^∞ manifolds without corners proven since page 1028 still holds for C^∞ manifolds with corners.

7/18/2026

Firstly, I realized that $(\mathbb{R}_{\geq 0})^n$ is way better notation than $\overline{\mathbb{R}_+}^n$. So I'm going to write the corner space in my way instead of John Lee's way. Anyways, before I get back to progressing through chapter 3 of Lee's book, I want to define one more standard manifold structure.

Suppose V is a finite dimensioned real vector space. Then there is a unique Hausdorff vector space topology on V (see [pages 442-443](#)). Furthermore, if $f : \mathbb{R}^n \rightarrow V$ is any invertible linear map and V is equipped with the unique Hausdorff topology, then f will be a homeomorphism. Therefore, can trivially say that any finite dimensioned real vector space is a topological manifold (without a boundary).

The standard smooth structure we equip V is the one generated by all the charts (V, f) where $f : V \rightarrow \mathbb{R}^n$ is a bijective linear map. Importantly, given any bijective linear maps $f, g : V \rightarrow \mathbb{R}^n$ we have that $g \circ f^{-1}$ is a bijective linear map from $\mathbb{R}^n$ to $\mathbb{R}^n$ (which is thus smooth). So, these charts do define a C^∞ -structure.

Suppose V is an n -dimensional real vector space equipped with the standard smooth structure. Then let $p \in V$. Given any $v \in V$ we define $D_v|_p : C^\infty(V, \mathbb{R}) \rightarrow \mathbb{R}$ by

$$D_v|_p f := \left. \frac{d}{dt} \right|_{t=0} f(p + tv).$$

First I want to show $D_v|_p$ is even well-defined. So suppose $f \in C^\infty(V)$ and let $S : \mathbb{R}^n \rightarrow V$ be any invertible linear map. Then we must have that $f \circ S \in C^\infty(\mathbb{R}^n)$.

Yet also let $a, y \in \mathbb{R}^n$ be such that $S(a) = p$ and $S(y) = v$. Then because:

$$S(a + ty) = S(a) + tS(y) = p + tv,$$

we can say that $f(p + tv) = (f \circ S)(a + ty)$.

It follows that $D_v|_p f = \nabla(f \circ S)(a) \cdot y = \nabla(f \circ S)(S^{-1}(p)) \cdot S^{-1}(v)$.

And based on the formula I just derived, it's also now clear that $D_v|_p$ is a derivation at p and that the map $V \rightarrow T_p V$ given by $v \mapsto D_v|_p$ is a linear map.

As a side note, if $V = \mathbb{R}^n$ then $D_v|_p$ is just the directional derivative at p in the direction of v .

Lemma: Suppose V is an n -dimensional real vector space and W is an m -dimensional real vector space. Also equip V and W with their standard smooth structure. Then any linear map $T : V \rightarrow W$ is a C^∞ map.

Why?

Let $S_V : \mathbb{R}^n \rightarrow V$ and $S_W : \mathbb{R}^m \rightarrow W$ be bijective linear maps. Then (V, S_V^{-1}) is a smooth chart on V covering all of V , (W, S_W^{-1}) is a smooth chart on W covering all of W , and $S_W^{-1} \circ T \circ S_V$ is a linear map (and thus a C^∞ -map) from $\mathbb{R}^n$ to $\mathbb{R}^m$. ■

Corollary: Let V be an n -dimensional real vector space equipped with the standard smooth structure, and also let V^* denote the "algebraic dual space" of V (i.e. the vector space of all linear functionals $\phi : V \rightarrow \mathbb{R}$ that are continuous or not continuous. [Although since V is finite-dimensional, all linear functionals will be continuous...]). Then $V^* \subseteq C^\infty(V, \mathbb{R})$.

Why?

The standard vector space smooth structure on $\mathbb{R}$ as well as the standard smooth structure on $\mathbb{R}$ are both generated by the chart $(\mathbb{R}, \text{Id}_{\mathbb{R}})$. Hence, both smooth structures are the same and we can in turn apply the prior lemma. ■

Proposition 3.13: Let V be an n -dimensional real vector spaces equipped with the standard smooth structure and suppose $p \in V$. The map $v \mapsto D_v|_p$ is a vector space isomorphism from V to $T_p V$.

Proof:

To start off, since V is an n -dimensional manifold, we know that $T_p V$ is also an n -dimensional vector space. Hence, so long as we can prove that the linear map $v \mapsto D_v|_p$ is injective then we must have that $v \mapsto D_v|_p$ is a vector space isomorphism.

Fortunately, suppose for the sake of contradiction that there exists $v \in V - \{0\}$ such that $D_v|_p f = 0$ for all $f \in C^\infty(V)$. Then we can pick vectors $v_2, \dots, v_n \in V$ such that $\text{span}(v, v_2, \dots, v_n) = V$. Next, there's a well-defined linear functional $f : V \rightarrow \mathbb{R}$ given by $f(c_1 v + \sum_{i=2}^n c_i v_i) := c_1$. Importantly, $f \in C^\infty(V, \mathbb{R})$ by the prior corollary. Also since $C^\infty(V, \mathbb{R})$ is a real vector space containing all constant functions, we can additionally say that $g := f - f(p)$ is a C^∞ map from V to $\mathbb{R}$.

With that, we now have that:

$$0 = D_v|_p g = \left. \frac{d}{dt} \right|_{t=0} g(p + tv) = \left. \frac{d}{dt} \right|_{t=0} f(tv) = \left. \frac{d}{dt} \right|_{t=0} t = 1.$$

This is a contradiction. So we can conclude that $\ker(v \mapsto D_v|_p) = \{0\}$.

Furthermore, suppose W is an m -dimensional real vector space equipped with the standard smooth structure and $L : V \rightarrow W$ is a linear map. Then for any $p \in V$ the below diagram commutes:

$$\begin{array}{ccccc}
 v & \xrightarrow{\quad} & D_v|_p & & \\
 \downarrow & & \searrow & & \\
 & V \xrightarrow{\sim} T_p V & & & \\
 L \downarrow & & \downarrow dL_p & & \\
 & W \xrightarrow{\sim} T_{L(p)} W & & & \\
 \downarrow & & \searrow dL_p(D_v|_p) & & \\
 L(w) & \xrightarrow{\quad} & D_{L(v)}|_{L(p)} & &
 \end{array}$$

Proof:

It suffices to show that $dL_p(D_v|_p) = D_{L(v)}|_{L(p)}$. Fortunately, for any $f \in C^\infty(W, \mathbb{R})$ we have that:

$$\begin{aligned}
 dL_p(D_v|_p)f &= D_v|_p(f \circ L) = \left. \frac{d}{dt} \right|_{t=0} (f \circ L)(p + tv) \\
 &= \left. \frac{d}{dt} \right|_{t=0} f(L(p) + tL(v)) = D_{L(v)}|_{L(p)}f. \blacksquare
 \end{aligned}$$

One may notice that the definition of a derivation on a C^∞ manifold M which was given prior specifically required us to work with maps in $C^\infty(M, \mathbb{R})$ maps. Also, we only defined differentials of C^∞ maps between maps. But what if we want to work with less smooth manifolds.

To start off, I want to recall a comment I made on [page 1035](#). Namely, suppose we are working with C^r manifold M (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$). Then given any $p \in M$ and $k < \min(\infty, r + 1)$, what is the space of "derivations at p " mapping $C^k(M, \mathbb{R})$ to $\mathbb{R}$?

I'm only slightly modifying a proof that I read from math stack exchange (see [citation 11](#) in my bibliography).

Recall that $C^k(M, \mathbb{R})$ is a unital real algebra. So, let us consider the set:

$$\mathfrak{m} := \{f \in C^k(M, \mathbb{R}) : f(p) = 0\}.$$

I claim this is a maximal ideal of $C^k(M, \mathbb{R})$.

To see that this is an ideal, just note that if $f \in \mathfrak{m}$ and $g \in C^k(M, \mathbb{R})$ then $fg \in C^k(M, \mathbb{R})$ with $(fg)(p) = 0 \cdot g(p) = 0$. So, $fg \in \mathfrak{m}$. Similarly, if $f, g \in \mathfrak{m}$ then $f + g \in C^k(M, \mathbb{R})$ with $(f + g)(p) = f(p) + g(p) = 0 + 0 = 0$. So, $f + g \in \mathfrak{m}$.

As for showing that $\mathfrak{m}$ is a maximal ideal, suppose $g \in C^k(M, \mathbb{R}) - \mathfrak{m}$. Then

consider any $f \in C^k(M, \mathbb{R})$. Since $g(p) \neq 0$, we can say that $f - \frac{f(p)}{g(p)}g \in \mathfrak{m}$. In turn, $f \in \mathbb{R}g + \mathfrak{m}$. And since f was arbitrary, we've proven that:

$$C^k(M, \mathbb{R}) = \mathbb{R}g + \mathfrak{m} \subseteq \langle \{g\} \cup \mathfrak{m} \rangle \subseteq C^k(M, \mathbb{R}).$$

Also since g was arbitrary, we've shown that $\mathfrak{m}$ is a maximal ideal (and even a maximal proper vector subspace) of $C^k(M, \mathbb{R})$.

Now $\mathfrak{m}^2$ will be an ideal (and thus also vector subspace) of $C^k(M, \mathbb{R})$ contained in $\mathfrak{m}$. Hence, we can consider the quotient space $V := \mathfrak{m}/\mathfrak{m}^2$. I claim that the space of "derivations at p " mapping $C^k(M, \mathbb{R})$ to $\mathbb{R}$ is isomorphic to the algebraic dual space V^* .

Proof:

Firstly suppose $\varphi \in V^* = (\mathfrak{m}/\mathfrak{m}^2)^*$. Equivalently, we can view φ as linear functional on $\mathfrak{m}$ such that $\varphi(\mathfrak{m}^2) = \{0\}$. Next, we know from before that $C^k(M, \mathbb{R}) = \{\text{constant functions}\} \oplus \mathfrak{m}$. So, we can uniquely extend φ to being a linear functional on $C^k(M, \mathbb{R})$ by just setting $\varphi(\text{constant function}) := 0$. For convenience, I'll give this extended linear functional the name D_φ .

I claim D_φ is a derivation at p . After all, we already know it's a linear map from $C^k(M, \mathbb{R})$ to $\mathbb{R}$. Furthermore, suppose $f, g \in C^k(M, \mathbb{R})$. Then we know that $\tilde{f} := f - f(p)$ and $\tilde{g} := g - g(p)$ are both in $\mathfrak{m}$. Also:

$$\begin{aligned} D_\varphi(fg) &= D_\varphi((\tilde{f} + f(p))(\tilde{g} + g(p))) \\ &= D_\varphi(\tilde{f}\tilde{g}) + f(p)D_\varphi(\tilde{g}) + g(p)D_\varphi(\tilde{f}) + D_\varphi(f(p)g(p)). \end{aligned}$$

But now as $D_\varphi(\mathfrak{m}^2) = \{0\}$, we know that $D_\varphi(\tilde{f}\tilde{g}) = 0$. Also, $D_\varphi(f(p)g(p)) = 0$, $D_\varphi(g - g(p)) = D_\varphi(g)$, and $D_\varphi(f - f(p)) = D_\varphi(f)$ since D_φ is a linear map sending constant functions to zero. Hence, we can conclude that:

$$D_\varphi(fg) = f(p)D_\varphi(g) + g(p)D_\varphi(f).$$

Now after letting $\mathcal{D}_p M$ denote the space of derivations at p mapping $C^k(M, \mathbb{R})$ to $\mathbb{R}$ define the map $\Phi : V^* \rightarrow \mathcal{D}_p M$ by $\varphi \mapsto D_\varphi$. It is obvious from the construction of D_φ that Φ is a linear map. Meanwhile, to show that Φ is bijective we shall note that Φ has an incredibly simple inverse.

Suppose $w \in \mathcal{D}_p M$ and consider $w|_{\mathfrak{m}}$. Since w is a derivation, we know from the lemma on [page 1028](#) that $w(\mathfrak{m}^2) = \{0\}$. [Technically that lemma assumes all the functions being input into the derivation are C^∞ . That said, nothing in the proof of that lemma actually requires any smoothness conditions.] It follows that $w|_{\mathfrak{m}}$ quotients to a well defined map $\Psi(w) \in (\mathfrak{m}^2/\mathfrak{m})^*$. Also, we can easily see that $\Psi = \Phi^{-1}$. ■

You may note that none of our reasoning in this section so far has required that k is finite. However, a fact one can prove is that if k is finite then V will be infinite dimensional. In turn, we'll also have that V^* is infinite dimensional. I'm tired, it's almost 1am, and proving this fact isn't the most insanely interesting thing ever. So, I'm going to skip showing this. But the T.L.D.R. is that if $k < \infty$ then there are way more derivations from $C^k(M, \mathbb{R})$ into $\mathbb{R}$ than there should be if derivations are purely an abstraction of directional derivatives. And we don't want this.

7/19/2026

If our goal is to have the tangent space to a C^r -manifold M (where r is potentially finite) only contain directional derivatives, then we can try to formalize a tangent space in the following different way from how we were doing so before. (Also for this section I'll be assuming by default that our manifolds can have corners and a boundary.)

Pick a point $p \in M$. Next, we consider any C^1 path $\gamma : J \rightarrow M$ where $J \subseteq \mathbb{R}$ is a closed interval containing 0 and having a nonempty interior, and $\gamma(0) = p$.

As a side note, I'm allowing 0 to be either endpoint of J .

Furthermore, as J is a 1-dimensional manifold (with a boundary) it does make sense to talk about γ as being C^1 or not. In fact, given any smooth chart (U, φ) on M , we can say that γ is a C^1 map if and only if γ is continuous and $\varphi \circ \gamma$ is a C^1 map from $\gamma^{-1}(U) \subseteq \mathbb{R}$.

We shall consider the set $\mathcal{C}(M, p)$ of all such C^1 -paths γ described priorly. And then, we define an equivalence relation on $\mathcal{C}(M, p)$. Namely, we say $\gamma_1 \sim \gamma_2$ iff:

$$(f \circ \gamma_1)'(0) = (f \circ \gamma_2)'(0) \text{ for all } f \in C^1(M, \mathbb{R}).$$

The fact that this is an equivalence relation is trivial to show. Hence, we can consider the space $\mathcal{V}_p M$ of equivalence classes of $\mathcal{C}(M, p)$. Plus, if $\gamma \in \mathcal{C}(M, p)$ then we consider its equivalence class in $\mathcal{V}_p M$ to be $[\gamma]$.

Lemma: Suppose $F : M \rightarrow N$ is a C^1 map and $\gamma_1 : J_1 \rightarrow M$ and $\gamma_2 : J_2 \rightarrow M$ are paths in $\mathcal{C}(M, p)$ such that $[\gamma_1] = [\gamma_2]$.

(a) We know $(F \circ \gamma_i) : J_i \rightarrow N$ is a path in $\mathcal{C}(N, F(p))$.

Hopefully this is obvious. After all, $F \circ \gamma_i$ is a composition of C^1 maps and is thus a C^1 map itself. Also $(F \circ \gamma_i)(0) = F(p)$.

(b) Furthermore, $[F \circ \gamma_1] = [F \circ \gamma_2]$ in $\mathcal{V}_{F(p)} N$.

Why?

Suppose $g \in C^1(N, \mathbb{R})$. Then $g \circ F \in C^1(M, \mathbb{R})$. Hence, we know that:

$$((g \circ F) \circ \gamma_1)'(0) = ((g \circ F) \circ \gamma_2)'(0). \blacksquare$$

Based on this, we'll define the differential at p of a C^1 map $F : M \rightarrow N$ to be the function $dF_p : \mathcal{V}_p M \rightarrow \mathcal{V}_{F(p)} N$ given by $dF_p[\gamma] := [F \circ \gamma]$.

(Ignore the conflict in notation now. I'll deal with it later...)

Proposition: Let M, N, P be C^r manifolds, let $F : M \rightarrow N$ and $G : N \rightarrow P$ be C^1 maps, and let $p \in M$.

(a) $d(G \circ F)_p = dG_{F(p)} \circ dF_p$.

Why?

$$d(G \circ F)_p([\gamma]) = [G \circ F \circ \gamma] = dG_{F(p)}([F \circ \gamma]) = dG_{F(p)}(dF_p([\gamma])).$$

(b) $d(\text{Id}_M)_p = \text{Id}_{\mathcal{V}_p M}$.

Hopefully this is obvious from the fact $[\text{Id}_M \circ \gamma] = [\gamma]$.

(c) If $F : M \rightarrow N$ is a C^1 -diffeomorphism then dF_p is a bijection and $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

To see why, just combine parts (a) and (b). ■.

Now a downside of viewing the tangent space $\mathcal{V}_p M$ as a collection of equivalence classes of curves is that the vector space structure on $\mathcal{V}_p M$ is not at all obvious. So my next goal will be to make that structure clear.

Lemma: Let M be an n -dimensional C^r manifold, let $p \in M$, and let (U, φ) be any smooth chart on M containing p . Then $\gamma_1, \gamma_2 \in \mathcal{C}(M, p)$ satisfy that $[\gamma_1] = [\gamma_2]$ if and only if $(\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0)$.

Proof:

Note for any $f \in C^1(M, \mathbb{R})$ that:

$$(f \circ \gamma_i)'(0) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (\varphi \circ \gamma_i)'(0)$$

Since $\nabla(f \circ \varphi^{-1})(\varphi(p))$ depends only on f and not on γ_i , it's now clear that if $(\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0)$ then we'll also have that $(f \circ \gamma_1)'(0) = (f \circ \gamma_2)'(0)$.

Conversely, suppose $v_1 := (\varphi \circ \gamma_1)'(0) \neq (\varphi \circ \gamma_2)'(0) =: v_2$. Then it suffices to find some $f \in C^1(M, \mathbb{R})$ such that:

$$\nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot v_1 \neq \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot v_2$$

Fortunately, because $v_1 \neq v_2$ in $\mathbb{R}^n$, we know there is some $w \in \mathbb{R}^n$ with $w \cdot v_1 \neq w \cdot v_2$. Next we can define $g : \mathbb{R}^n \rightarrow \mathbb{R}$ by $g(x) = x \cdot w$. Clearly g is a C^∞ function on $\varphi(U)$ with $\nabla g(\varphi(p)) = w$. If we then further set $\tilde{g} := g \circ \varphi$, we have that $\tilde{g} \in C^r(M, \mathbb{R})$.

By standard arguments on LCH spaces as well as with bump functions on C^r manifolds, we can find a precompact open set $V \subseteq M$ with $p \in V \subseteq \overline{V} \subseteq U$ as well as a function $f \in C^r(M, \mathbb{R}) \subseteq C^1(M, \mathbb{R})$ such that $f = \tilde{g}$ on $\overline{V}$. And at last, we have on a neighborhood of $\varphi(p)$ that:

$$\begin{aligned} \nabla(f \circ \varphi^{-1})(\varphi(p)) &= \nabla(\tilde{g} \circ \varphi^{-1})(\varphi(p)) \\ &= \nabla(g \circ \varphi \circ \varphi^{-1})(\varphi(p)) = \nabla g(\varphi(p)) = w. \blacksquare \end{aligned}$$

One consequence of the prior lemma is that for any smooth chart (U, φ) on an n -dimensional manifold M containing a point p , there is a well-defined injective map $T_\varphi : \mathcal{V}_p M \rightarrow \mathbb{R}^n$ given by $T_\varphi([\gamma]) := (\varphi \circ \gamma)'(0)$.

Proposition: Suppose p is *not* a corner point of M . Then the map T_φ defined above is also surjective.

Proof:

Consider any $w \in \mathbb{R}^n$ and note that since $\varphi(U)$ must be an open neighborhood of $\varphi(p)$ in either H^n or $\mathbb{R}^n$, we know for a small enough $R > 0$ that either $\varphi(p) + tw \in \varphi(U)$ or $\varphi(p) - tw \in \varphi(U)$ for all $t \in [0, R]$.

Now we can define a map γ going from either $[0, R]$ or $[-R, 0]$ to M by setting $\gamma(t) := \varphi^{-1}(\varphi(p) + tw)$. In either case, we have that $\gamma \in \mathcal{C}(M, p)$ and:

$$(\varphi \circ \gamma)'(0) = \left. \frac{d}{dt} \right|_{t=0} (\varphi(p) + tw) = w. \blacksquare$$

Unfortunately, this proof doesn't work if p is a corner point of M . In fact, if p is a corner point of M then we'll not even have that $T_\varphi(\mathcal{T}_p M)$ is a vector subspace of $\mathbb{R}^n$. So from here on out in this section I'll be assuming my C^r manifolds do not have corners.

Now that we have a bijection $T_\varphi : \mathcal{V}_p M \rightarrow \mathbb{R}^n$, we can let this bijection induce a vector space structure on $\mathcal{V}_p M$. In other words, given any $[\gamma_1], [\gamma_2] \in \mathcal{V}_p M$ as well as any $c_1, c_2 \in \mathbb{R}$, we define:

$$c_1[\gamma_1] + c_2[\gamma_2] := T_\varphi^{-1}(c_1 T_\varphi([\gamma_1]) + c_2 T_\varphi([\gamma_2])).$$

Proposition: The vector space structure on $\mathcal{V}_p M$ is defined independently of any particular smooth chart (U, φ) on M containing p . In particular, suppose (V, ψ) is another smooth chart on M containing p . Then for any $[\gamma_1], [\gamma_2] \in \mathcal{V}_p M$ as well as any $c_1, c_2 \in \mathbb{R}$ we have that:

$$T_\varphi^{-1}(c_1 T_\varphi([\gamma_1]) + c_2 T_\varphi([\gamma_2])) = T_\psi^{-1}(c_1 T_\psi([\gamma_1]) + c_2 T_\psi([\gamma_2]))$$

Proof:

Define the vectors:

- $v := c_1 T_\varphi([\gamma_1]) + c_2 T_\varphi([\gamma_2]) = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0),$
- $w := c_1 T_\psi([\gamma_1]) + c_2 T_\psi([\gamma_2]) = c_1(\psi \circ \gamma_1)'(0) + c_2(\psi \circ \gamma_2)'(0).$

Next let $[\tilde{\gamma}] := T_\varphi^{-1}(v)$. Thus $\tilde{\gamma}$ is a C^1 path on M with $\tilde{\gamma}(0) = p$ and $T_\varphi([\tilde{\gamma}]) = (\varphi \circ \tilde{\gamma})'(0) = v$. To prove the proposition, we need to show that $[\tilde{\gamma}] = T_\psi^{-1}(w)$. Or in other words, we need to show that $T_\psi([\tilde{\gamma}]) = (\psi \circ \tilde{\gamma})'(0) = w$.

Fortunately, $\tilde{\gamma}$ will map some open neighborhood of 0 in the interval $\tilde{\gamma}$ is defined on into the set $U \cap V$. Therefore, we can say by chain rule that:

$$(\psi \circ \tilde{\gamma})'(0) = (\psi \circ \varphi^{-1} \circ \varphi \circ \tilde{\gamma})'(0) = [(D(\psi \circ \varphi^{-1}))(\varphi(p))] (\varphi \circ \tilde{\gamma})'(0)$$

Next note that the matrix $[(D(\psi \circ \varphi^{-1}))(\varphi(p))]$ defines a linear map. Therefore as $(\varphi \circ \tilde{\gamma})'(0) = v = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0)$, we can say that:

$$\begin{aligned} & [(D(\psi \circ \varphi^{-1}))(\varphi(p))] (\varphi \circ \tilde{\gamma})'(0) \\ &= c_1 [(D(\psi \circ \varphi^{-1}))(\varphi(p))] (\varphi \circ \gamma_1)'(0) + c_2 [(D(\psi \circ \varphi^{-1}))(\varphi(p))] (\varphi \circ \gamma_2)'(0) \end{aligned}$$

Finally, by one more application of chain rule (which we can use on γ_1 and γ_2 for identical reasons as why we could use it on $\tilde{\gamma}$) we get that:

$$[(D(\psi \circ \varphi^{-1}))(\varphi(p))] (\varphi \circ \gamma_i)'(0) = (\psi \circ \gamma_i)'(0).$$

Thus, we've proven that $(\psi \circ \tilde{\gamma})'(0) = c_1(\psi \circ \gamma_1)'(0) + c_2(\psi \circ \gamma_2)'(0) = w$. $\blacksquare$

Now that $\mathcal{V}_p M$ has been equipped with a standard vector space structure, I've now almost shown everything I want to on this tangent. What's left to do is show how $\mathcal{V}_p M$ corresponds to the space of derivations at p on M .

Lemma: Suppose M and N are C^r manifolds and $F : M \rightarrow N$ is a C^1 map. Then for any $p \in M$ we have that $dF_p : \mathcal{V}_p M \rightarrow \mathcal{V}_{F(p)} N$ is a linear map.

Proof:

Consider any $c_1, c_2 \in \mathbb{R}$ as well as $\gamma_1, \gamma_2 \in \mathcal{C}(M, p)$. Next let $\tilde{\gamma} \in \mathcal{C}(M, p)$ be such that $[\tilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$ in $\mathcal{V}_p M$. To prove this lemma it suffices to show that:

$$[F \circ \tilde{\gamma}] = dF_p[\tilde{\gamma}] = c_1 dF_p[\gamma_1] + c_2 dF_p[\gamma_2] = c_1[F \circ \gamma_1] + c_2[F \circ \gamma_2]$$

in $\mathcal{V}_{F(p)} N$.

But now because $F \in C^1(M, N)$ we know there is a smooth chart (U, φ) on M containing p as well as a smooth chart (V, ψ) on N containing $F(p)$ such that $F(U) \subseteq V$ and $\psi \circ F \circ \varphi^{-1}$ is a C^1 map. Based on how we defined the vector space structure for $\mathcal{V}_{F(p)} N$, we just need to show that:

$$(\psi \circ F \circ \tilde{\gamma})'(0) = c_1(\psi \circ F \circ \gamma_1)'(0) + c_2(\psi \circ F \circ \gamma_2)'(0).$$

Fortunately, we also know since $[\tilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$ that:

$$(\varphi \circ \tilde{\gamma})'(0) = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0).$$

And now if we let $\mathbf{A}$ denote the matrix $(D(\psi \circ F \circ \varphi^{-1}))(\varphi(p))$, we have by chain rule that:

$$\begin{aligned} (\psi \circ F \circ \tilde{\gamma})'(0) &= \mathbf{A}(\varphi \circ \tilde{\gamma})'(0) = c_1 \mathbf{A}(\varphi \circ \gamma_1)'(0) + c_2 \mathbf{A}(\varphi \circ \gamma_2)'(0) \\ &= c_1(\psi \circ F \circ \gamma_1)'(0) + c_2(\psi \circ F \circ \gamma_2)'(0). \blacksquare \end{aligned}$$

Lemma: Let M be a C^r manifold and let $\mathcal{D}_p M$ denote the space of derivations at $p \in M$ mapping $C^k(M, \mathbb{R})$ to $\mathbb{R}$ (where $k \leq r$). Then there is a well-defined map $\Psi : \mathcal{V}_p M \rightarrow \mathcal{D}_p M$ given by $\Psi([\gamma])f := (f \circ \gamma)'(0)$.

Why?

The fact that $\Psi([\gamma])$ is a well-defined injective function on $C^1(M, \mathbb{R})$ is obvious from the definition of the equivalence class of γ . As for showing that $\Psi([\gamma])$ is a derivation, just apply normal derivative rules to $((c_1 f + c_2 g) \circ \gamma)'$ and $((fg) \circ \gamma)'$.

Furthermore, Ψ is a linear map.

Why?

Fix any $f \in C^1(M, \mathbb{R})$ and smooth chart (U, φ) on M containing p , and recall from the proof of the lemma on [page 1047](#) that:

$$(f \circ \gamma)'(0) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (\varphi \circ \gamma)'(0) \text{ for all } \gamma \in \mathcal{C}(M, p).$$

But now if $[\tilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$, we know that:

$$(\varphi \circ \tilde{\gamma})'(0) = c_1(\varphi \circ \gamma_1)'(0) + c_2(\varphi \circ \gamma_2)'(0).$$

Since $x \mapsto \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot x$ is a linear map, we can now easily conclude that if $[\tilde{\gamma}] = c_1[\gamma_1] + c_2[\gamma_2]$ then $(f \circ \gamma)'(0) = c_1(f \circ \gamma_1)'(0) + c_2(f \circ \gamma_2)'(0)$.

Plus I claim no matter what k is, Ψ is still injective.

When $k = 1$ then this is just a result of the definition of the equivalence classes in $\mathcal{V}_p M$. After all, $[\gamma_1] \neq [\gamma_2]$ if and only if there exists some function $f \in C^1(M, \mathbb{R})$ with $(f \circ \gamma_1)'(0) \neq (f \circ \gamma_2)'(0)$. Meanwhile, if $1 < k \leq r$ then the thing we need to specifically show is that if $[\gamma_1] \neq [\gamma_2]$ then we can more strongly take f to be a

C^k map (as opposed to just a C^1 map) such that $(f \circ \gamma_1)'(0) \neq (f \circ \gamma_2)'(0)$. Since $C^r(M, \mathbb{R}) \subseteq C^k(M, \mathbb{R})$ for all $k \leq r$, I might as well by default just look for some $f \in C^r(M, \mathbb{R})$. Fortunately, I already showed how to construct such a function $f \in C^r(M, \mathbb{R})$ in my proof of the lemma on [page 1047](#). ■

Proposition: Let M, N be C^r manifolds and suppose $F : M \rightarrow N$ is a C^k map (where $k \leq r$). Given any $p \in M$ we let $\mathcal{D}_p M$ denote the space of derivations at p from $C^k(M, \mathbb{R})$ to $\mathbb{R}$. Similarly, we let $\mathcal{D}_{F(p)} N$ denote the space of derivations at $F(p)$ from $C^k(N, \mathbb{R})$ to $\mathbb{R}$. And finally, we let $\hat{d}F_p : \mathcal{D}_p M \rightarrow \mathcal{D}_{F(p)} N$ be the linear map $\hat{d}F_p(v)f = v(f \circ F)$. Then the following diagram commutes:

$$\begin{array}{ccc}
 [\gamma] & \xrightarrow{\quad} & f \mapsto (f \circ \gamma)'(0) \\
 \downarrow & \begin{array}{ccc} \mathcal{V}_p M & \hookrightarrow & \mathcal{D}_p M \\ \downarrow dF_p & & \downarrow \hat{d}F_p \\ \mathcal{V}_{F(p)} N & \hookrightarrow & \mathcal{D}_{F(p)} N \end{array} & \downarrow \\
 [F \circ \gamma] & \xrightarrow{\quad} & g \mapsto (g \circ F \circ \gamma)'(0)
 \end{array}$$

Hopefully this is obvious. Also as a side note, if $k = \infty$ then the horizontal arrows must be vector space isomorphisms. After all, $\mathcal{V}_p M$ and $\mathcal{D}_p M$ will both be n -dimensional vector spaces (assuming M is an n -dimensional manifold), and we just showed the arrow $\mathcal{V}_p M \rightarrow \mathcal{D}_p M$ is given by an injective linear map.

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Derivations are much nicer to work with than equivalence classes of curves. So I guess the take away from all the prior reasoning in this tangent is that we now have a formal way of describing which derivations we care about (i.e. which derivations actually correspond to "directional derivatives"). And importantly, this reasoning works even for non- C^∞ -manifolds.

So if M is a C^r manifold without corners and $\Psi : \mathcal{V}_p M \rightarrow \mathcal{D}_p M$ is the function from the lemma on the prior page, then we let $T_p M := \Psi(\mathcal{V}_p M)$.
